COMPLETE EXAM CHEAT SHEET — ONE DAY PREP

SECTION 1: LOGICAL REASONING

ARRANGEMENTS

Linear Arrangement — Key Rules

- Read all clues first, find the **definite position** clues first
- Use a table/grid: positions as columns, people as rows
- "A is 3rd from left" = fixed anchor → build around it
- "A is to the immediate left of B" = A B are adjacent
- "Exactly 2 people between A and B" = count gaps, not positions

Circular Arrangement — Key Rules

- **Facing centre:** Left = anticlockwise, Right = clockwise
- **Facing outside:** Left = clockwise, Right = anticlockwise
- Fix ONE person first → place others relatively
- "A is 2nd to the right of B" → count clockwise from B

Matrix Arrangement

- Two variables (e.g., floors + flats, rows + columns)
- Make a 2D grid and eliminate systematically

PUZZLES

Floor-Based Puzzle

- N floors: Ground = 1st, Top = Nth
- "Above/Below" clues → use number line
- "Immediately above/below" = adjacent floors

Box-Based Puzzle

- Box 1 = bottom, Box N = top (usually)
- Treat exactly like floor puzzles

Scheduling Puzzles

- Make a table: Days × People/Tasks
- Fix definite days first, then apply gap/order clues

BLOOD RELATIONS

Relation	Meaning
Father's brother	Uncle
Mother's brother	Maternal Uncle
Brother's son	Nephew
Sister's daughter	Niece
Father's father	Grandfather
Spouse's father	Father-in-law

Shortcut: Draw a family tree. Male = △, Female = ○ **Coded relations:** Decode step by step, left to right.

CODING-DECODING

Type	Trick
Letter shift	Find the shift value (A→D = +3)
Reverse coding	Z=1, Y=2... or word reversed
Number coding	Position of letter (A=1, B=2...)
Symbol coding	Consistent pattern across examples

Golden rule: Always compare 2 coded examples to find the pattern.

DIRECTION SENSE

```
      NORTH
        ↑
WEST ←——[YOU]——→ EAST
        ↓
      SOUTH
```

- After turning **Right** → rotate 90° clockwise
- After turning **Left** → rotate 90° anticlockwise
- Shadow in morning → West side | Shadow in evening → East side
- **Distance formula** for displacement: $\sqrt{(a^2 + b^2)}$

INPUT-OUTPUT

- Find the **rule applied in each step** (shift, sort, swap)
- Common patterns: words arranged alphabetically, numbers ascending/descending, alternating word-number
- Work backwards from output if asked for an intermediate step

---

SYLLOGISMS

All + All = All | All + No = No | Some + All = Some Some + No = Some Not | No + No = No conclusion

Venn Diagram Method (Fastest)

- Draw circles for each term
- Check if conclusion **must be true** in ALL possible diagrams
- **Possibility questions:** Check if conclusion CAN be true in any diagram

Key Traps

- "Some A are B" → does NOT mean "Some B are A" is false (it's actually true)
- "All A are B" → Some A are B ✅ | Some B are A ✅ (both valid conclusions)

STATEMENTS & CONCLUSIONS / CAUSE & EFFECT

Statements & Conclusions:

- Conclusion must **logically follow** from the statement alone

- Don't use outside knowledge
- "Definitely true" = must follow | "Probably true" = may follow

Cause & Effect:

- Cause happens **before** effect
- Both must be **related**, not coincidental
- Independent causes → can both cause the same effect

SERIES

Number Series — Common Patterns

Pattern	Example
+n, +n	2, 5, 8, 11 (+3)
×n	2, 6, 18, 54 (×3)
n^2	1, 4, 9, 16, 25
$n^2 + k$	2, 5, 10, 17 (n^2+1)
Difference series	Differences themselves form a series
Alternating	Two interleaved series

Alphabet Series

- A=1, B=2 ... Z=26 (memorise this)
- EJOTY trick: **E=5, J=10, O=15, T=20, Y=25**
- Find gap between letters, check if pattern is consistent

DATA SUFFICIENCY

Answer choices (standard):

- A → Statement 1 alone sufficient
- B → Statement 2 alone sufficient
- C → Both together sufficient
- D → Either alone sufficient
- E → Both together not sufficient

Trick: Don't solve fully — just check **if it CAN be solved**, not the answer itself.

SECTION 2: ABSTRACT REASONING

Series of Figures

- Check: Size, Shape, Rotation, Shading, Number of elements, Position
- Changes are usually **consistent** per step

Odd One Out

- Find the property that **4 share** and 1 doesn't
- Check: Number of sides, symmetry, shading, rotation direction

Mirror Images

- **Vertical mirror (left-right flip):** Left ↔ Right, Top stays
- **Horizontal mirror (top-bottom flip):** Top ↔ Bottom, Left stays
- Letters flipped: b↔d, p↔q, M stays, A stays

Water Images

- Top-bottom flip (upside down)
- Letters flipped: n↔u, b↔p, d↔q

Paper Folding

- Unfold in reverse — holes mirror across fold line
- Fold once = 1 line of symmetry for holes

Cubes & Dice

Opposite faces of a standard die: 1-6, 2-5, 3-4

- Two faces that share an edge can **never** be opposite
- If same face visible in two positions → find the common face to orient

Embedded Figures

- Rotate the inner figure mentally
- Look for a figure where all lines of the shape exist within the larger figure

SECTION 3: QUANTITATIVE APTITUDE

ARITHMETIC SHORTCUTS

Percentages

Fraction	%
1/2	50%
1/3	33.33%
1/4	25%
1/5	20%
1/6	16.67%
1/8	12.5%
1/10	10%
1/12	8.33%

- % increase = (Increase/Original) × 100
- If price up by 20% → to restore, decrease by **16.67%** (not 20%)
- Successive % change: a% then b% → Net = a + b + ab/100

Profit & Loss

- Profit% = (Profit/CP) × 100
- SP = CP × (100 + P%)/100
- CP = SP × 100/(100 + P%)
- Dishonest trader formula: Profit% = (True weight - False weight)/False weight × 100

Time, Speed & Distance

- **Speed = Distance / Time**
- Same distance, speeds S1 & S2 → Average speed = **2S1S2 / (S1+S2)**
- Relative speed (same direction) = S1 − S2
- Relative speed (opposite direction) = S1 + S2

- Train crossing a pole: Time = Length of train / Speed
- Train crossing a platform: Time = (Train + Platform length) / Speed

---

Time & Work

- If A does work in **n days** → A's 1 day work = **1/n**
- A + B together: 1/A + 1/B = 1/Total days
- **Shortcut:** Total days = (A × B)/(A + B)
- If A is twice as efficient as B → A takes half the time

Ratio & Proportion

- If A:B = 3:5 and total = 80 → A = 30, B = 50
- Componendo: (a+b)/(a-b) when a/b is given
- Partnership: Profit split = Ratio of (Capital × Time)

Simple & Compound Interest

- **SI = P × R × T / 100**
- **CI = P(1 + R/100)^n − P**
- CI − SI for 2 years = P(R/100)^2
- CI − SI for 3 years = P(R/100)^2 × (R/100 + 3)

Averages

- Average = Sum / Count
- New average when number added: (Old sum + New number) / (Old count + 1)
- If average increases by x when one member added → New member = Old avg + x(n+1)

Mixtures & Alligations

```
Cheaper (C)        Dearer (D)
      \            /
       \          /
        Mean (M)
       /          \
      /            \
  (D - M)        (M - C)   ← Ratio to mix
```

ALGEBRA

Linear Equations

- Two equations, two variables → eliminate one variable
- Substitution or cross-multiplication method

Quadratic Equations

- $ax^2 + bx + c = 0$
- Roots: $x = [-b \pm \sqrt{(b^2-4ac)}] / 2a$
- Sum of roots = $-b/a$ | Product of roots = c/a
- If discriminant $(b^2-4ac) > 0$ → two real roots | $= 0$ → equal roots | < 0 → no real roots

---

GEOMETRY SHORTCUTS

Shape	Area	Perimeter
Square (a)	a^2	$4a$
Rectangle (l,b)	$l \times b$	$2(l+b)$
Triangle (b,h)	$\frac{1}{2} \times b \times h$	sum of sides
Circle (r)	πr^2	$2\pi r$
Trapezium	$\frac{1}{2}(a+b) \times h$	—

Solid	Volume	Surface Area
Cube (a)	a^3	$6a^2$
Cuboid	$l \times b \times h$	$2(lb+bh+lh)$
Cylinder (r,h)	$\pi r^2 h$	$2\pi r(r+h)$
Cone (r,h)	$\frac{1}{3}\pi r^2 h$	$\pi r(r+l)$
Sphere (r)	$4/3\pi r^3$	$4\pi r^2$

Triangle properties:

- Sum of angles = 180°
- Pythagoras: $a^2 + b^2 = c^2$

- Special triangles: 3-4-5, 5-12-13, 8-15-17

PERMUTATION & COMBINATION

- **nPr = n! / (n−r)!** → Ordered arrangement
- **nCr = n! / [r!(n−r)!]** → Selection (order doesn't matter)
- nCr = nC(n−r)
- Circular arrangement of n items = **(n−1)!**
- Arrangement with identical items: n! / (p! × q!)

PROBABILITY

- P(Event) = Favourable outcomes / Total outcomes
- P(A') = 1 − P(A)
- P(A or B) = P(A) + P(B) − P(A and B)
- P(A and B) = P(A) × P(B) — only if **independent**
- Cards: 52 total | 4 suits | 13 each suit | 12 face cards

DATA INTERPRETATION — QUICK APPROACH

1. **Read the title and units first** (thousands? lakhs? %)
2. **Don't calculate everything** — only what's asked
3. For % change: (New − Old)/Old × 100
4. For ratio questions: simplify before calculating
5. Approximate when options are far apart (saves time)

SECTION 4: VERBAL ABILITY & READING COMPREHENSION

READING COMPREHENSION — STRATEGY

Question Type	Approach
Main Idea	Read 1st + last paragraph carefully

Inference	Must be **implied**, not stated directly
Tone/Attitude	Look for positive/negative/neutral words
Title	Should cover the **whole** passage, not one part
Vocabulary in context	Read full sentence around the word

Speed tip: Read the **questions first**, then the passage — saves re-reading.

VOCABULARY

Common Synonyms & Antonyms

Word	Synonym	Antonym
Benevolent	Kind, Generous	Malevolent
Candid	Frank, Honest	Evasive
Verbose	Wordy, Lengthy	Concise
Obscure	Vague, Hidden	Clear, Obvious
Prudent	Wise, Careful	Reckless
Lethargic	Sluggish, Lazy	Energetic
Acrimonious	Bitter, Harsh	Cordial
Frugal	Thrifty	Extravagant
Ephemeral	Short-lived	Permanent
Audacious	Bold, Daring	Timid
Placid	Calm, Peaceful	Agitated
Sagacious	Wise, Shrewd	Foolish

GRAMMAR QUICK RULES

Tenses (Most Tested)

Tense	Structure	Signal words

Simple Present	V1 / V1+s	always, usually, every day
Present Continuous	is/am/are + V-ing	now, currently, at present
Simple Past	V2	yesterday, ago, last year
Past Perfect	had + V3	before, by the time, already
Future Perfect	will have + V3	by tomorrow, by next year

Subject-Verb Agreement

- Singular subject → singular verb (He **runs**)
- Either/Neither/Each/Every → singular verb
- Collective nouns (team, committee) → usually singular
- "A number of" → plural | "The number of" → singular

Articles

- **a** = before consonant sound (a university ✅ — 'yu' sound)
- **an** = before vowel sound (an hour ✅ — 'ou' sound)
- **the** = specific, already known, unique things
- No article: languages, sports, meals (generally)

Prepositions

Use	Preposition
Time (specific)	at (at 5pm, at noon)
Time (days/dates)	on (on Monday, on 5th)
Time (months/years)	in (in June, in 2024)
Place (point)	at
Place (surface)	on
Place (inside)	in

PARA JUMBLES — METHOD

1. Find the **opening sentence** — introduces topic, no pronoun without reference, no connector
2. Find the **closing sentence** — conclusion, summary, or consequence
3. Look for **connector words:** However, Therefore, Moreover, Thus, But, Although
4. Look for **pronoun references** — "he/she/it/they" must follow the noun they refer to
5. Arrange pairs first, then build the full sequence

ERROR DETECTION — COMMON ERRORS

Error Type	Wrong	Correct
Subject-verb	He don't know	He doesn't know
Double comparative	More better	Better
Wrong preposition	Married with	Married to
Redundancy	Return back	Return
Article error	She is honest girl	She is an honest girl
Plural with each	Each students	Each student
Than/Then	She is taller then him	than him

SENTENCE COMPLETION — TIPS

- Look for **contrast clues** (but, although, however → opposite meaning needed)
- Look for **similarity clues** (and, also, similarly → same meaning needed)
- Eliminate options that don't fit the **tone** of the sentence
- For two-blank questions, eliminate by testing the **easier blank** first